

Foxconn Open BMC 新人訓

L1 - Introduction to BMC features

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環境設定

系統資訊

項目	內容
作業系統	Ubuntu 22.04.5 LTS (VirtualBox)
核心版本	6.8.0-87-generic
Python	3.10.12
硬碟空間	1TB(約 750GB 可用)

OpenBMC 環境

項目	內容
OpenBMC 版本	2.18.0 (commit: 05f7a3f2ed)
MACHINE	romulus
原始碼路徑	~/openbmc
Build 路徑	~/openbmc/build/romulus
映像檔路徑	~/openbmc/build/romulus/tmp/deploy/images/romulus/

筆試測驗

作業目標

完成 L1_quiz.sh 筆試測驗，測驗包含 10 題單選題和 10 題複選題，測試對 BMC 基礎知識的理解。

步驟 1: 執行筆試測驗

目的: 透過互動式測驗，驗證對 OpenBMC 架構、BitBake、D-Bus 等基礎概念的掌握程度。

```
cd ~/Newcomer_training/L1/L1_quiz  
./L1_quiz.sh
```



```
=====
🏆 測驗結束：工號 835051
答對題數：20 / 20
總分      : 100 / 100
=====
```

重點說明：

- 測驗涵蓋：BMC 功能、Yocto/BitBake 建置系統、D-Bus 通訊機制
- 單選題 10 題 + 複選題 10 題，需全部答對才能得滿分

✓ 筆試測驗 完成：20/20 題答對，總分 100/100 分。

Exercise 1 - C++ 程式與 Systemd 服務

作業目標

在 meta-bmc-training 專案下創建 C++ 程式，設定為 systemd service，並在 QEMU 中驗證執行。程式需每 5 秒輸出一次：

BMC L1 Training - ID : 835051, Name : Sonny Chu

項目	內容
程式檔案	bmc_L1_ID835051.cpp
Service 檔案	bmc.L1.id835051.service
執行檔位置	/usr/bin/bmc-l1-hello
驗證環境	QEMU romulus

步驟 1: 檢視專案結構

目的: 確認 meta-bmc-training 專案的檔案結構，了解 BitBake recipe 的組織方式。

```
tree ~/openbmc/meta-bmc-training/recipes-L1/
```

```
sonny@sonny-VirtualBox:~/Newcomer_training/L1/L1_quiz$ tree ~/openbmc/meta-bmc-training/recipes-L1/
~/openbmc/meta-bmc-training/recipes-L1/
├── bmc-l1-hello
│   ├── bmc-l1-hello_1.0.bb
│   └── files
│       ├── bmc_L1_ID835051.cpp
│       └── bmc.L1.id835051.service
2 directories, 3 files
```

重點說明：

- bmc-l1-hello/: Recipe 資料夾，名稱對應 BitBake recipe
- bmc-l1-hello_1.0.bb: BitBake recipe 檔案，定義如何編譯和安裝
- files/: 存放原始碼和設定檔的資料夾

步驟 2: 檢視 C++ 程式碼

目的: 確認程式邏輯: 使用無限迴圈每 5 秒輸出一次訓練訊息，適合作為背景服務。

```
cat
```

```
~/openbmc/meta-bmc-training/recipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp
```

```
sonny@sonny-VirtualBox:~/Newcomer_training/L1/L1_quiz$ cat ~/openbmc/meta-bmc-training/recipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp
#include <iostream>
#include <unistd.h>

int main() {
    while (true) {
        std::cout << "BMC L1 Training - ID : 835051, Name : Sonny Chu" << std::endl;
        sleep(5);
    }
    return 0;
}
// Test modification
// Another test
```

重點說明：

- while(true): 無限迴圈, 讓程式持續運行不會結束
- sleep(5): 每次輸出後暫停 5 秒, 避免 CPU 過度使用
- std::cout: 輸出到標準輸出, 可被 systemd 的 journal 記錄

步驟 3: 檢視 Systemd Service 設定檔

目的: 確認 service 設定: 開機自動啟動, 異常時自動重啟。

```
cat
~/openbmc/meta-bmc-training/recipes-l1/bmc-l1-hello/files/bmc.L1.id835051.service
```

```
sonny@sonny-VirtualBox:~/Newcomer_training/L1/L1_quiz$ cat ~/openbmc/meta-bmc-training/recipes-l1/bmc-l1-hello/files/bmc.L1.id835051.service
[Unit]
Description=BMC L1 Training Service - ID 835051
After=multi-user.target

[Service]
ExecStart=/usr/bin/bmc-l1-hello
Restart=always

[Install]
WantedBy=multi-user.target
```

重點說明:

- [Unit] After=multi-user.target: 等系統進入多使用者模式後才啟動
- [Service] Restart=always: 程式異常結束時, systemd 會自動重新啟動
- [Install] WantedBy=multi-user.target: 設定開機時自動啟動

步驟 4: 驗證服務運行狀態

目的: 在 QEMU 環境中確認服務正常運行, 並檢視輸出訊息。

```
systemctl status bmc.L1.id835051.service
```

```
root@romulus:~# systemctl status bmc.L1.id835051.service
* bmc.L1.id835051.service - BMC L1 Training Service - ID 835051
   Loaded: loaded (/usr/lib/systemd/system/bmc.L1.id835051.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-12-04 02:00:30 UTC; 1h 35min ago
   Invocation: b8d0ab354fc042f9a9ce112bdc0a01c5
   Main PID: 373 (bmc-l1-hello)
    CPU: 835ms
   CGroup: /system.slice/bmc.L1.id835051.service
           └─373 /usr/bin/bmc-l1-hello

Dec 04 03:35:03 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:08 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:13 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:18 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:23 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:28 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:33 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:38 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:43 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Dec 04 03:35:48 romulus bmc-l1-hello[373]: BMC L1 Training - ID : 835051, ...Chu
Hint: Some lines were ellipsized, use -l to show in full.
root@romulus:~#
```

重點說明:

- Active: active (running): 表示服務正在運行中
- Main PID: 373: 服務的主程序 ID
- 日誌顯示每 5 秒輸出一行「BMC L1 Training - ID : 835051」

✓ **Exercise 1 完成:** 成功創建 C++ 程式並設定為 systemd service, 在 QEMU 中驗證運行正常。

Exercise 2 - Git 版本控制操作

作業目標

練習 Git 版本控制的常用操作，包含暫存、提交、分支切換、stash 暫存等功能，這些是日常開發必備的技能。

步驟 1: git add 與 git commit (暫存與提交)

目的: 將修改的檔案加入暫存區，然後提交到本地倉庫，建立版本記錄。

```
echo "// L1 test" >> recipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp
git status
git add .
git commit -m "Add L1 test modification"
```

```
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ echo "// L1 test" >> recipes
-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git status
On branch bmc_L1_ID835051
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   recipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp

no changes added to commit (use "git add" and/or "git commit -a")
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git add .
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git commit -m "Add L1 test m
odification"
[bmc_L1_ID835051 c020aa5] Add L1 test modification
 1 file changed, 1 insertion(+)
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$
```

重點說明:

- git status: 顯示工作目錄狀態，列出已修改但未暫存的檔案(紅色)
- git add .: 將所有變更加入暫存區(staging area)，準備提交
- git commit -m: 提交暫存區的變更，-m 參數指定提交訊息

步驟 2: git checkout (分支切換)

目的: 在不同分支間切換，每個分支可以有獨立的開發進度。

```
git checkout master
git branch
git checkout bmc_L1_ID835051
git branch
```

```
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git checkout master
Switched to branch 'master'
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git branch
  bmc_L1_ID835051
  bmc_L2_ID835051
* master
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git checkout bmc_L1_ID835051
Switched to branch 'bmc_L1_ID835051'
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git branch
* bmc_L1_ID835051
  bmc_L2_ID835051
  master
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$
```

重點說明:

- `git checkout <branch>`: 切換到指定分支
- `git branch`: 列出所有本地分支, * 號標記目前所在分支
- 用途: 不同功能在不同分支開發, 避免互相影響

步驟 3: `git stash` (暫存變更)

目的: 臨時儲存未提交的修改, 讓工作目錄回到乾淨狀態, 方便切換分支處理其他事情。

```
echo "// Another test" >> recipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp
git status
git stash
git status
```

```
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ echo "// Another test" >> re
cipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git status
On branch bmc_L1_ID835051
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   recipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp

no changes added to commit (use "git add" and/or "git commit -a")
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git stash
Saved working directory and index state WIP on bmc_L1_ID835051: c020aa5 Add L1 t
est modification
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git status
On branch bmc_L1_ID835051
nothing to commit, working tree clean
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$
```

重點說明:

- 先用 `echo` 指令故意製造一個未提交的修改
- `git stash`: 將目前的修改暫存起來, 工作目錄變回乾淨狀態
- 使用情境: 開發到一半需要緊急修 bug, 先 `stash` 再切換分支

步驟 4: `git stash pop` (恢復變更)

目的: 從 `stash` 清單中取回之前暫存的修改, 繼續未完成的工作。

```
git stash list
git stash pop
git status
```

```

sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git stash list
stash@{0}: WIP on bmc_L1_ID835051: c020aa5 Add L1 test modification
stash@{1}: WIP on bmc_L1_ID835051: c020aa5 Add L1 test modification
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git stash pop
On branch bmc_L1_ID835051
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   recipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (265ba36add177e8d629ccb48211da4f112d82ecf)
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git status
On branch bmc_L1_ID835051
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   recipes-L1/bmc-l1-hello/files/bmc_L1_ID835051.cpp

no changes added to commit (use "git add" and/or "git commit -a")
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$

```

重點說明：

- git stash list: 列出所有暫存的變更, stash@{0} 是最新的
- git stash pop: 取出最新的暫存並套用, 同時從清單中移除
- git stash apply: 與 pop 類似, 但不會從清單中移除(可重複使用)

步驟 5: git fetch / pull / push (遠端操作)

目的: 與遠端倉庫同步: 下載更新、合併變更、推送本地提交。

```

git fetch
git pull
git push

```

```

sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git fetch
sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git pull
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

If you wish to set tracking information for this branch you can do so with:

    git branch --set-upstream-to=<remote>/<branch> bmc_L1_ID835051

sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$ git push
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository using

    git remote add <name> <url>

and then push using the remote name

    git push <name>

sonny@sonny-VirtualBox:~/openbmc/meta-bmc-training$

```

重點說明：

- git fetch: 從遠端下載最新資訊, 但不合併到本地分支
- git pull: 下載並合併到本地(相當於 fetch + merge)
- git push: 將本地提交推送到遠端倉庫

備註: 因為沒有設定遠端倉庫, 所以顯示錯誤訊息。實際專案中需先用 git remote add 設定遠端。

✓ **Exercise 2 完成:** 熟悉 Git 版本控制操作, 包含 add、commit、checkout、stash 等日常開發必備指令。

Exercise 3 - IPMI 操作

作業目標

使用 ipmitool 指令取得 SDR (Sensor Data Record) 資訊, 分別以 In-band 和 Out-of-band 兩種方式測試, 了解兩者的差異與應用場景。

步驟 1: 編譯並部署 ipmitool

目的: romulus 參考平台預設未包含 ipmitool, 需要自行編譯並部署到 QEMU 環境。

備註: romulus 是社群參考平台, 映像檔較精簡。實際產品 (如 Intel 平台) 通常已內建 ipmitool。

```
cd ~/openbmc
```

```
source setup romulus
```

```
bitbake ipmitool
```

```
sonny@sonny-VirtualBox:~/openbmc/build/romulus$ cd ~/openbmc
sonny@sonny-VirtualBox:~/openbmc$ source setup romulus
Machine romulus found in meta-ibm/meta-romulus
Common targets are:
  obmc-phosphor-image
sonny@sonny-VirtualBox:~/openbmc/build/romulus$ bitbake ipmitool

Initialising tasks: 100% |#####| Time: 0:00:01
NOTE: Executing Tasks
NOTE: Tasks Summary: Attempted 1554 tasks of which 1535 didn't need to be rerun
and all succeeded.
sonny@sonny-VirtualBox:~/openbmc/build/romulus$
```

編譯完成後, 使用 scp 傳送到 QEMU:

```
scp -P 2222 ~/openbmc/build/romulus/tmp/work/.../ipmitool
root@127.0.0.1:/usr/bin/
```

重點說明:

- bitbake ipmitool: 使用 sstate-cache, 編譯很快 (約 1 分鐘)
- scp -P 2222: 透過 SSH 端口轉發傳檔案到 QEMU

步驟 2: In-band 測試 (帶內測試)

目的: 在 BMC 內部直接執行 ipmitool, 不需要網路連線。

```
ipmitool sdr list
```

```

root@romulus:~# ipmitool sdr list
IANA PEN registry open failed: No such file or directory
Failed to load entries from IANA PEN Registry
BootProgress      | 0x00          | ok
occ0              | disabled     | ns
occ1              | disabled     | ns
cpu0              | disabled     | ns
cpu1              | disabled     | ns
dimm0             | disabled     | ns
dimm1             | disabled     | ns
dimm2             | disabled     | ns
dimm3             | disabled     | ns
dimm4             | disabled     | ns
dimm5             | disabled     | ns
dimm6             | disabled     | ns
dimm7             | disabled     | ns
dimm8             | disabled     | ns
dimm9             | disabled     | ns
dimm10            | disabled     | ns
dimm11            | disabled     | ns
dimm12            | disabled     | ns
dimm13            | disabled     | ns
dimm14            | disabled     | ns
dimm15            | disabled     | ns
dimm0_temp        | disabled     | ns
dimm1_temp        | disabled     | ns
dimm2_temp        | disabled     | ns
dimm3_temp        | disabled     | ns
dimm4_temp        | disabled     | ns
dimm5_temp        | disabled     | ns
dimm6_temp        | disabled     | ns
dimm7_temp        | disabled     | ns
dimm8_temp        | disabled     | ns
dimm9_temp        | disabled     | ns
dimm10_temp       | disabled     | ns
dimm11_temp       | disabled     | ns
dimm12_temp       | disabled     | ns
dimm13_temp       | disabled     | ns
dimm14_temp       | disabled     | ns
dimm15_temp       | disabled     | ns
cpu0_core0        | disabled     | ns
cpu0_core1        | disabled     | ns
cpu0_core2        | disabled     | ns
cpu0_core3        | disabled     | ns
cpu0_core4        | disabled     | ns
cpu0_core5        | disabled     | ns

```

cpu0_core6	disabled	ns
cpu0_core7	disabled	ns
cpu0_core8	disabled	ns
cpu0_core9	disabled	ns
cpu0_core10	disabled	ns
cpu0_core11	disabled	ns
cpu0_core12	disabled	ns
cpu0_core13	disabled	ns
cpu0_core14	disabled	ns
cpu0_core15	disabled	ns
cpu0_core16	disabled	ns
cpu0_core17	disabled	ns
cpu0_core18	disabled	ns
cpu0_core19	disabled	ns
cpu0_core20	disabled	ns
cpu0_core21	disabled	ns
cpu0_core22	disabled	ns
cpu0_core23	disabled	ns
cpu1_core0	disabled	ns
cpu1_core1	disabled	ns
cpu1_core2	disabled	ns
cpu1_core3	disabled	ns
cpu1_core4	disabled	ns
cpu1_core5	disabled	ns
cpu1_core6	disabled	ns
cpu1_core7	disabled	ns
cpu1_core8	disabled	ns
cpu1_core9	disabled	ns
cpu1_core10	disabled	ns
cpu1_core11	disabled	ns
cpu1_core12	disabled	ns
cpu1_core13	disabled	ns
cpu1_core14	disabled	ns
cpu1_core15	disabled	ns
cpu1_core16	disabled	ns
cpu1_core17	disabled	ns
cpu1_core18	disabled	ns
cpu1_core19	disabled	ns
cpu1_core20	disabled	ns
cpu1_core21	disabled	ns
cpu1_core22	disabled	ns
cpu1_core23	disabled	ns
p0_core0_temp	disabled	ns
p0_core1_temp	disabled	ns
p0_core2_temp	disabled	ns

```

p0_core3_temp | disabled | ns
p0_core4_temp | disabled | ns
p0_core5_temp | disabled | ns
p0_core6_temp | disabled | ns
p0_core7_temp | disabled | ns
p0_core8_temp | disabled | ns
p0_core9_temp | disabled | ns
p0_core10_temp | disabled | ns
p0_core11_temp | disabled | ns
p0_core12_temp | disabled | ns
p0_core13_temp | disabled | ns
p0_core14_temp | disabled | ns
p0_core15_temp | disabled | ns
p0_core16_temp | disabled | ns
p0_core17_temp | disabled | ns
p0_core18_temp | disabled | ns
p0_core19_temp | disabled | ns
p0_core20_temp | disabled | ns
p0_core21_temp | disabled | ns
p0_core22_temp | disabled | ns
p0_core23_temp | disabled | ns
p1_core0_temp | disabled | ns
p1_core1_temp | disabled | ns
p1_core2_temp | disabled | ns
p1_core3_temp | disabled | ns
p1_core4_temp | disabled | ns
p1_core5_temp | disabled | ns
p1_core6_temp | disabled | ns
p1_core7_temp | disabled | ns
p1_core8_temp | disabled | ns
p1_core9_temp | disabled | ns
p1_core10_temp | disabled | ns
p1_core11_temp | disabled | ns
p1_core12_temp | disabled | ns
p1_core13_temp | disabled | ns
p1_core14_temp | disabled | ns
p1_core15_temp | disabled | ns
p1_core16_temp | disabled | ns
p1_core17_temp | disabled | ns
p1_core18_temp | disabled | ns
p1_core19_temp | disabled | ns
p1_core20_temp | disabled | ns
p1_core21_temp | disabled | ns
p1_core22_temp | disabled | ns
p1_core23_temp | disabled | ns
AttemptsLeft | 0x00 | ok
OperatingSystemS | 0x00 | ok
root@romulus:~#

```

重點說明：

- In-band: 在 BMC 內部 (QEMU 終端) 直接執行
- 前兩行警告 (IANA PEN registry) 是正常的, 不影響功能
- romulus 的 sensor 大多顯示 disabled/ns, 因為是模擬環境沒有實際硬體

步驟 3: Out-of-band 測試 (帶外測試)

目的: 從遠端主機透過網路連接到 BMC, 這是伺服器管理最常用的方式。

```
ipmitool -I lanplus -C 17 -H 127.0.0.1 -p 6623 -U root -P OpenBmc sdr list
```

參數說明：

參數	說明
-I lanplus	使用 IPMI 2.0 加密協定 (LAN+)
-C 17	Cipher Suite 17 (HMAC-SHA256 加密)
-H 127.0.0.1	目標 BMC 的 IP 位址
-p 6623	IPMI 端口 (QEMU 轉發自 UDP 623)
-U root	BMC 使用者名稱

-P 0penBmc	BMC 密碼
------------	--------

```
sonny@sonny-VirtualBox:~/openbmc/build/romulus$ ipmitool -I lanplus -C 17 -H 127
.0.0.1 -p 6623 -U root -P 0penBmc sdr list
BootProgress      | 0x00          | ok
occ0               | disabled      | ns
occ1               | disabled      | ns
cpu0               | disabled      | ns
cpu1               | disabled      | ns
dimm0              | disabled      | ns
dimm1              | disabled      | ns
dimm2              | disabled      | ns
dimm3              | disabled      | ns
dimm4              | disabled      | ns
dimm5              | disabled      | ns
dimm6              | disabled      | ns
dimm7              | disabled      | ns
dimm8              | disabled      | ns
dimm9              | disabled      | ns
dimm10             | disabled      | ns
dimm11             | disabled      | ns
dimm12             | disabled      | ns
dimm13             | disabled      | ns
dimm14             | disabled      | ns
dimm15             | disabled      | ns
dimm0_temp         | disabled      | ns
dimm1_temp         | disabled      | ns
dimm2_temp         | disabled      | ns
dimm3_temp         | disabled      | ns
dimm4_temp         | disabled      | ns
dimm5_temp         | disabled      | ns
dimm6_temp         | disabled      | ns
dimm7_temp         | disabled      | ns
dimm8_temp         | disabled      | ns
dimm9_temp         | disabled      | ns
dimm10_temp        | disabled      | ns
dimm11_temp        | disabled      | ns
dimm12_temp        | disabled      | ns
dimm13_temp        | disabled      | ns
dimm14_temp        | disabled      | ns
dimm15_temp        | disabled      | ns
cpu0_core0         | disabled      | ns
cpu0_core1         | disabled      | ns
cpu0_core2         | disabled      | ns
cpu0_core3         | disabled      | ns
cpu0_core4         | disabled      | ns
cpu0_core5         | disabled      | ns
cpu0_core6         | disabled      | ns
cpu0_core7         | disabled      | ns
```

cpu0_core11	disabled	ns
cpu0_core12	disabled	ns
cpu0_core13	disabled	ns
cpu0_core14	disabled	ns
cpu0_core15	disabled	ns
cpu0_core16	disabled	ns
cpu0_core17	disabled	ns
cpu0_core18	disabled	ns
cpu0_core19	disabled	ns
cpu0_core20	disabled	ns
cpu0_core21	disabled	ns
cpu0_core22	disabled	ns
cpu0_core23	disabled	ns
cpu1_core0	disabled	ns
cpu1_core1	disabled	ns
cpu1_core2	disabled	ns
cpu1_core3	disabled	ns
cpu1_core4	disabled	ns
cpu1_core5	disabled	ns
cpu1_core6	disabled	ns
cpu1_core7	disabled	ns
cpu1_core8	disabled	ns
cpu1_core9	disabled	ns
cpu1_core10	disabled	ns
cpu1_core11	disabled	ns
cpu1_core12	disabled	ns
cpu1_core13	disabled	ns
cpu1_core14	disabled	ns
cpu1_core15	disabled	ns
cpu1_core16	disabled	ns
cpu1_core17	disabled	ns
cpu1_core18	disabled	ns
cpu1_core19	disabled	ns
cpu1_core20	disabled	ns
cpu1_core21	disabled	ns
cpu1_core22	disabled	ns
cpu1_core23	disabled	ns
p0_core0_temp	disabled	ns
p0_core1_temp	disabled	ns
p0_core2_temp	disabled	ns
p0_core3_temp	disabled	ns
p0_core4_temp	disabled	ns
p0_core5_temp	disabled	ns
p0_core6_temp	disabled	ns
p0_core7_temp	disabled	ns
p0_core8_temp	disabled	ns
p0_core9_temp	disabled	ns

```

p0_core9_temp | disabled | ns
p0_core10_temp | disabled | ns
p0_core11_temp | disabled | ns
p0_core12_temp | disabled | ns
p0_core13_temp | disabled | ns
p0_core14_temp | disabled | ns
p0_core15_temp | disabled | ns
p0_core16_temp | disabled | ns
p0_core17_temp | disabled | ns
p0_core18_temp | disabled | ns
p0_core19_temp | disabled | ns
p0_core20_temp | disabled | ns
p0_core21_temp | disabled | ns
p0_core22_temp | disabled | ns
p0_core23_temp | disabled | ns
p1_core0_temp | disabled | ns
p1_core1_temp | disabled | ns
p1_core2_temp | disabled | ns
p1_core3_temp | disabled | ns
p1_core4_temp | disabled | ns
p1_core5_temp | disabled | ns
p1_core6_temp | disabled | ns
p1_core7_temp | disabled | ns
p1_core8_temp | disabled | ns
p1_core9_temp | disabled | ns
p1_core10_temp | disabled | ns
p1_core11_temp | disabled | ns
p1_core12_temp | disabled | ns
p1_core13_temp | disabled | ns
p1_core14_temp | disabled | ns
p1_core15_temp | disabled | ns
p1_core16_temp | disabled | ns
p1_core17_temp | disabled | ns
p1_core18_temp | disabled | ns
p1_core19_temp | disabled | ns
p1_core20_temp | disabled | ns
p1_core21_temp | disabled | ns
p1_core22_temp | disabled | ns
p1_core23_temp | disabled | ns
AttemptsLeft | 0x00 | ok
OperatingSystemS | 0x00 | ok
sonny@sonny-VirtualBox:~/openbmc/build/romulus$

```

In-band vs Out-of-band 比較

項目	In-band (帶內)	Out-of-band (帶外)
執行位置	BMC 內部	遠端主機
連線方式	本地直接執行	網路連線 (UDP 623)
需要網路	否	是
使用場景	BMC 本地診斷、除錯	遠端監控、大規模管理
安全性	需要 BMC 存取權限	需要 IPMI 帳號密碼

✓ **Exercise 3 完成:** 完成 IPMI In-band 和 Out-of-band 測試, 了解兩種方式的差異與應用場景。

結論

完成項目總結

項目	內容	結果
筆試測驗	BMC 基礎知識測驗	100/100 分
Exercise 1	C++ 程式 + Systemd 服務	✓ 完成
Exercise 2	Git 版本控制操作	✓ 完成
Exercise 3	IPMI In-band / Out-of-band	✓ 完成

學習心得

透過本次 L1 訓練，對 OpenBMC 的基礎架構有了初步認識：

- BitBake/Yocto 建置系統：了解如何創建 recipe、編譯套件、建立映像檔
- Systemd 服務管理：學會創建和管理背景服務
- Git 版本控制：掌握日常開發必備的版本管理技能
- IPMI 協定：理解 BMC 遠端管理的基礎，區分 In-band 和 Out-of-band 的差異

這些知識為後續更進階的 L2-L13 訓練奠定了基礎。